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# Enhancing Flare System Efficiency Through Innovative Technologies

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## Abstract

The study presents a comprehensive analysis of the High Pressure (HP) Flare Header system, emphasizing its critical role in the safe management of hydrocarbon processing facilities. The study details the integration of the HP Flare Header with multiple upstream sources, addressing operational considerations such as back pressure issues due to flare tip design. It highlights the importance of maintaining system integrity under both routine and emergency conditions while ensuring compliance with industry safety and environmental standards. The analysis includes an overview of flare systems, their significance in safety and environmental protection, and the operational efficiency they provide. Special attention is given to sonic flares, their enhanced combustion efficiency, and the challenges associated with back pressure. The study proposes modifications to the existing flare tip design to mitigate back pressure issues, ensuring reliable performance and safety. The proposed modifications aim to improve back pressure management, enhance operational stability, and reduce greenhouse gas emissions, contributing to a more efficient and environmentally friendly flare system.

## Introduction

The safe management of hydrocarbon processing facilities relies heavily on effective pressure relief and venting systems. Central to this safety infrastructure is the High Pressure (HP) Flare Header system, which serves as a critical conduit for safely directing relief streams from various equipment—including blowdown valves (BDV), pressure safety valves (PSVs), vapour recovery units (VRUs), and other process control devices—towards controlled disposal or recovery points. Sonic flares are well known for smokeless combustion, reduce flame length and lower radiation. However higher back pressure in a sonic flare system can negatively impact the efficiency of a low-pressure system such as glycol regeneration unit. This is because the flare system's back pressure can increase the pressure in the reboiler, which is a key component of the glycol regeneration process. Higher reboiler pressure reduces the efficiency of water removal from the glycol, leading to lower glycol purity and potentially causing operational issues. The aim of this paper is to present a modified design of sonic tip that allows low back pressure during normal operating scenarios yet retaining sonic capabilities at higher flow.

## Flare System: An Overview

A flare system is an essential safety device employed in various industries, particularly in oil and gas, petrochemical, and chemical processing plants. Its primary function is to safely combust excess hydrocarbons and other gases that cannot be processed or recycled. This process is crucial in preventing the release of harmful gases into the atmosphere, thereby safeguarding both the environment and human health.

### Importance of Flare Systems

***Safety.*** The foremost importance of a flare system lies in its ability to handle emergency situations. In the event of equipment failure, power outages, or other operational issues, the flare system acts as a safety system, preventing the buildup of pressure that could lead to explosions or other hazardous situations. By providing a controlled means of disposing of excess gases, flare systems mitigate the risk of catastrophic incidents.

***Environmental Protection.*** Flare systems play a pivotal role in environmental protection by burning off excess gases, thereby reducing the emission of volatile organic compounds (VOCs) and other harmful pollutants. This is crucial for meeting environmental regulations and minimizing the impact of industrial operations on air quality. The combustion process in flare systems converts harmful gases into less harmful byproducts, such as carbon dioxide and water vapor, thus contributing to cleaner air.

***Operational Efficiency.*** Flare systems contribute to maintaining operational efficiency by safely disposing of excess gases. They also help prevent disruptions in the production process and ensure that the plant operates smoothly. This is particularly important in industries where continuous operation is critical to profitability and safety. The presence of a reliable flare system allows for safe handling of process upsets and maintenance activities, thereby enhancing overall operational stability and flexibility.

### Sonic Flare: An Introduction

A sonic flare, also known as a high-pressure flare, is a type of flare system designed to handle high-pressure gas streams. Unlike conventional flares, which operate at subsonic speeds, sonic flares operate at supersonic speeds, resulting in a more efficient and complete combustion process. The design of sonic flares incorporates advanced engineering principles to achieve optimal performance under high-pressure conditions.

### Importance of Sonic Flares

***Enhanced Combustion Efficiency.*** Sonic flares are capable of achieving higher combustion efficiency compared to conventional flares. This is due to the supersonic speed at which the gas is expelled, which promotes better mixing of the gas with air and results in a more complete combustion process. The high velocity of the gas stream ensures thorough mixing with the ambient air, leading to efficient oxidation of the hydrocarbons.

***Reduced Emissions.*** The improved combustion efficiency of sonic flares leads to lower emissions of unburned hydrocarbons and other pollutants. This makes them an attractive option for industries looking to minimize their environmental footprint. By achieving near-complete combustion. Sonic flares significantly reduce the release of harmful byproducts, contributing to cleaner air and compliance with stringent environmental regulations.

***Handling High-Pressure Gas Streams.*** Sonic flares are specifically designed to handle high-pressure gas streams, making them suitable for applications where conventional flares may not be effective and efficient. This includes situations where the gas stream contains a high concentration of hydrogen or other light hydrocarbons. The robust design of sonic flares allows them to withstand the demanding conditions associated with high-pressure operations, ensuring reliable performance and safety.

### Back Pressure Issues with Sonic Flares

While sonic flares offer several advantages, they are not without their challenges. One of the primary issues associated with sonic flares is back pressure. Back pressure refers to the resistance encountered by the gas as it flows through the flare system. High back pressure can lead to several problems, including:

***Reduced Combustion Efficiency.*** High back pressure can impede the flow of gas through the flare system, resulting in incomplete combustion and higher emissions of unburned hydrocarbons. The resistance to gas flow can disrupt the optimal mixing of the gas with air, leading to inefficient combustion and increased pollutant emissions.

***Increased Wear and Tear.*** The increased resistance caused by high back pressure can lead to greater wear and tear on the flare system components, reducing their lifespan and increasing maintenance costs. The mechanical stress imposed by high back pressure can accelerate the degradation of critical components, necessitating more frequent inspections and replacements.

***Operational Disruptions.*** High back pressure can also cause operational disruptions, as it may require the plant to reduce production rates or shut down certain processes and equipment to manage the pressure buildup. The need to address pressure issues can lead to unplanned downtime and reduced operational efficiency, impacting overall productivity and profitability.

### Mitigating Back Pressure Issues

To address the back pressure issues associated with sonic flares, several strategies can be employed:

***Proper Design and Sizing.*** Ensuring that the flare system is properly designed and sized for the specific application is crucial. This includes selecting the appropriate flare tip, stack height, and other components to minimize back pressure. A well-designed flare system considers the characteristics of the gas stream, operating conditions, and environmental factors to achieve optimal performance.

***Regular Maintenance.*** Regular maintenance of the flare system can help identify and address any issues that may contribute to back pressure. This includes inspecting and cleaning the flare tip, checking for blockages, and ensuring that all components are in good working condition. Proactive maintenance practices can prevent the buildup of deposits and other obstructions that can increase back pressure.

***Use of Pressure Relief Devices.*** Installing pressure relief devices, such as pressure relief valves or rupture discs, can help manage back pressure by allowing excess gas to be safely vented from the system. These devices provide a controlled means of relieving pressure, preventing the buildup of excessive back pressure and ensuring the safe operation of the flare system.

***Optimizing Operating Conditions.*** Adjusting the operating conditions of the flare system, such as the gas flow rate and pressure, can also help minimize back pressure and improve overall performance. By fine-tuning the operating parameters, operators can achieve a balance between efficient combustion and manageable back pressure, ensuring the reliable and safe operation of the flare system.

## Existing Sonic Flare Tip and Associated Problems

The current gas dehydration and injection facility is equipped with a flare system that utilizes a variable area flare tip. This tip results in relatively high back pressure even at lower flow rates, negatively impacting the operation of existing low-pressure systems such as the Glycol and Amine regeneration system, and the Vapor recovery unit during start-up or changeover processes.

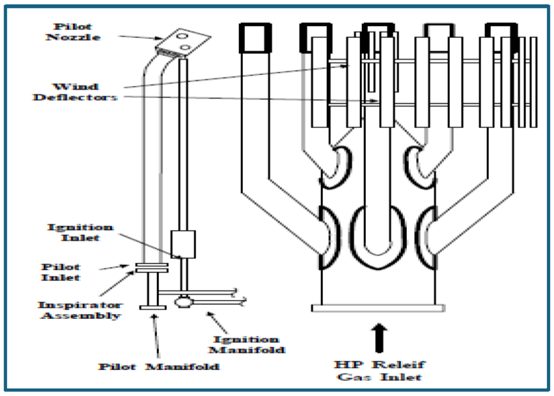


**Figure 1—Existing Sonic Flare-Tip Design**

Sonic flares rely on the exit velocity to entrain and mix large quantities of air, to provide smokeless, low heat radiation flaring. As the exit velocity increases, the ability to entrain the air also increases. The velocity and momentum also help to lift the waste gas flame off the flare, reducing or eliminating damaging flame lick, therefore increasing longevity of the flare tip. Flare size is a function of the maximum allowable backpressure vs maximum flowrate (along with gas composition and temperature). As flares get larger, turndown becomes more of a concern. Fixed exit area flares have very limited turndown while the variable area flare has an infinite turndown. But, for any maximum flow, backpressure is needed to achieve the velocity required.

The flare tip being sonic, offers relatively higher backpressure in the flare header even at lower flowrates (e.g. 12 psig for approx. 2.5 MMSCFD), however it has infinite turndown and does achieve smokeless, low heat radiation flaring for all flows from purge to peak. As per the design of the sonic flare, the nozzles are designed to lift at approximately 12 psig. The springs hold the insert in place until the velocity is maximized (sonic velocity) at 12 psig. Due to having higher backpressure than the original flare tip, some operational issues were observed in the plant.

The flare system in existing facility is comprised of collection headers, HP flare knockout drum, flare ignition system, subsea flare pipeline and remote flare tower with sonic flare tip.

The flare header collection system includes three (3) segregated headers as follows.

a. Wet header
b. Dry header
c. Liquid collection header

The wet header collects all relief, blow down and vent connections from process equipment and piping systems upstream of the gas injection compressors and from all utility systems. Refer to the figure below for the relief sources joining the header.

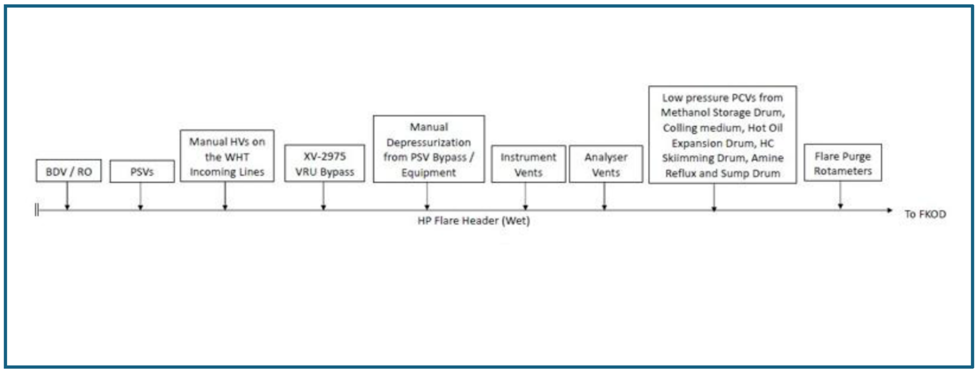


**Figure 2—Wet Flare Header Relief Source**

The dry header is dedicated to relief, blow down and vent connections resulting from high pressure dehydrated gas sources i.e. the injection gas compression system and injection gas distribution. Refer to the figure below for the relief sources joining the header.

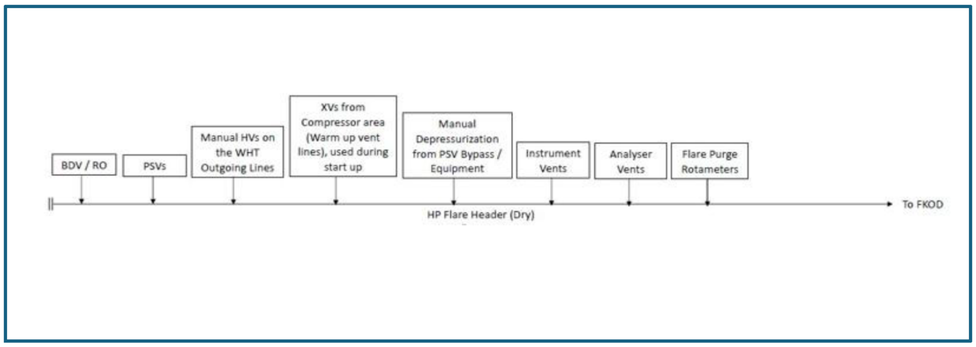


**Figure 3—Dry Flare Header Relief Source**

A dedicated liquid collection header is provided to collect any intermittent process drains. Refer to the figure below for the relief sources joining the header.

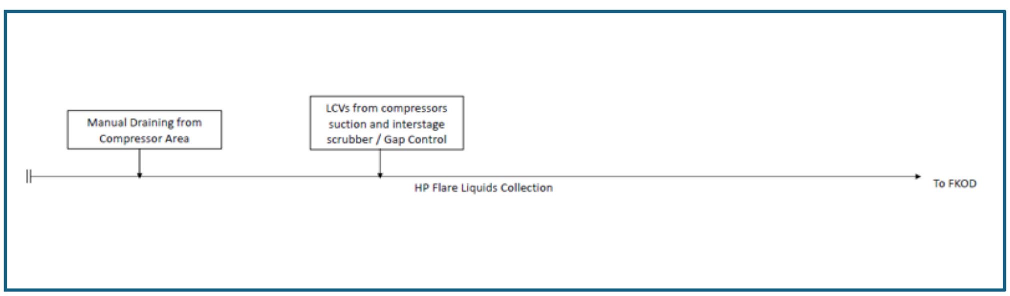


**Figure 4—Liquid Collection Header Relief Source**

The wet and dry flare collection headers are normally swept with nitrogen to prevent air ingress at the remote sonic flare tip. Low pressure fuel gas purge connections are also provided as back-up in the event of nitrogen unavailability.

Any liquids collected in the Flare KO drum is routed under gap control through to the Closed Drains Drum for onward disposal. Vapour from the FKOD is routed to the flare tower via subsea pipeline.

During normal operation, vapour from the closed drain drum along with comingled vapour from Gas Sweetening unit regeneration system and the Dehydration unit regeneration system are routed to the vapour recovery compressor. The compressed waste gas is exported via subsea pipeline.

A spill-over to the flare system is provided upstream of the vapour recovery unit (VRU) compressor system for upset conditions only, by opening the quick opening valve. When the VRU suction pressure/ flow is very high and can no longer be handled through VRU, the quick opening valve opens to route the gas to flare stack via KOD.

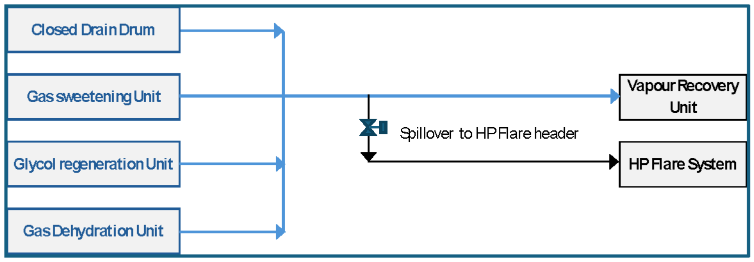


**Figure 5—VRU schematic**

## Operating Scenarios and Issues

### VRU Start-up / Change-over

Gas injection compressors (GIC) are started after making sure that process gas dew point & fuel gas H2S specifications are below acceptable limit. After start-up of GIC's and when enough acid gases are produced, VRU start-up is initiated. During this interim time, ~ 15.5 MMSCFD (Waste gas + Dilution gas) is flared. With the present flare tip, the back• pressure reaches above 12 psig in this case. But as VRU suction high-high pressure trip is set at 6.8 psig, proper start-up of VRU is prevented.

### VRU Outage

During the outage of VRU flare header back-pressure is expected to be around 12 psig. Any compressor shutdown at this point will further increase in back pressure and may reach higher than 20 psig resulting in tripping of Glycol dehydration unit.

### Glycol Regeneration Issues

High back pressure in flare header results in operation Glycol reboiler at elevated pressure with associated glycol purity issues and circulation of Glycol in regeneration flow loop.

### Amine Regeneration Issues

Amine regenerator operates on 14.5 psig operating pressure, when flare back-pressure is 12 psig, there is only 2 psi difference for Amine regenerator Acid gases to pass through to flare.

### Liquid during VRU Start-up

Currently, during VRU trip high flare backpressure is experienced which results in condensed liquid hold up in Waste gas KOD upstream pipe works. During restart of VRU, while compressors start to rotate, initial pulling of gas from waste gas system causes sudden dump of huge quantity of accumulated liquid from pipe works (hold up volume during high flare header back pressure) to Waste gas KOD which cannot be handled by this vessel in normal operating condition and may reach high-high level / plant trip scenario. Currently, a temporary procedure is in place to start VRU by mitigating all abnormal situations caused by FT-6 back pressure.

These operational issues prevent start-up of VRU post stoppage / changeover due to the increased backpressure, unless the dilution gas is stopped which is not advisable. Consequently, if VRU is not started, the flare backpressure will always remain high.

## Options to overcome the operational challenges

### Replacement of Sonic tip with Fixed pipe tip

The fixed pipe flares have a fixed exit area, meaning the flare tip does not change size or shape. They are sized for peak relief rates from process facilities. Fixed pipe flares will solve all the operational issues currently experienced in the plant, as these are designed with very low pressure drop at normal operating condition (2 psi up to approx. 100 MMSCFD).

However, in normal operation, relief rates are very low/velocity – comprising of mainly flare purge flows. When the gas exiting the flare tip has a low velocity, it's more susceptible to being affected by wind or other disturbances, potentially causing the flame to travel back into the flare instead of being contained at the tip. This could damage the flare tip / pilot due to flame lick / lazy flame (being a non-sonic flare). Fixed flares are also susceptible to flame-out, particularly during upset conditions, leading to the release of unburned hydrocarbons.

### Flare Tip Modification[1,2]

The existing flare tip has a provision of 10" spare connection (currently kept blinded), included as part of original design. This can be used to install a new low-pressure arm, such that the backpressure in the flare for low flow is reduced. Adding the low-pressure arm is significantly different than providing a single large area flare. It is true that the first stage arm is a fixed area, but it is small and controllable. The nozzle will be designed identically to a typical sonic nozzle to ensure longevity and can safely handle flows from purge (low pressure) to peak flow.

The initially proposed modification with additional low-pressure arm offered a backpressure of 6 psig for 20 MMSCFD flow.

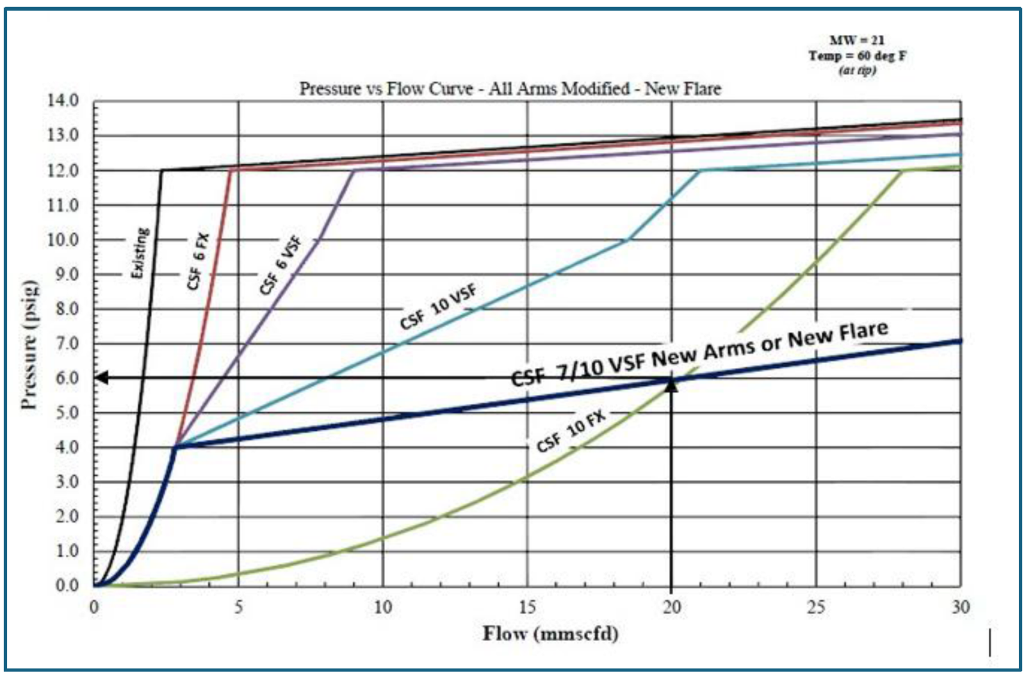


Figure 6—Pressure Vs Flow Curve – with additional low-pressure arm[2]

Since the VRU trip set point is 6.8 psig, and to flatten the back pressure curve beyond 20 MMSCFD the design was further optimized by replacing existing flare tip arms with low pressure opening (LP design).

These modified arms will still close under purge conditions but will open to provide added low flow area, at lower pressures. The two LP arms will allow for expanded capacity (beyond 20 MMSCFD) without significant increases in pressure.

These modifications[2] will turn the existing flare into a 3-stage system, without the use of any controls or staging valves. The system will react based on flow and pressure. Low flows will pass through the single arm (bypass arm). This is stage 1. As flows and pressure exceed 2 to 2.5 psig, the 2 LP arms will begin to lift, giving added low-pressure capacity. The CSF-10-FX arm and the two CSF-12-VSF LP arms represent stage 2. As flows and pressures exceed 12 psig, all arms would be in service and represent the third stage and handle all capacity to the maximum flow. Following is the summary of the different stages mentioned above.

a. Stage – 1: 10" Bypass Arm (Primary Flame, post modification, 0 to 2 psig)
b. Stage – 2: CSF-VSF modified arms (LP Design on the modified flow arms, 2 to 12 psig)
c. Stage – 3: Rest of the existing Flare (12 psig and above).

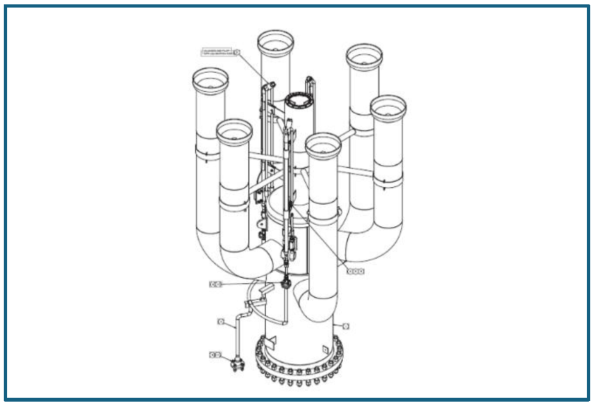

Figure 7—New Flare tip design[1]

The flare header backpressure vs flow curve for the above option along with 10" arm modification is as below.

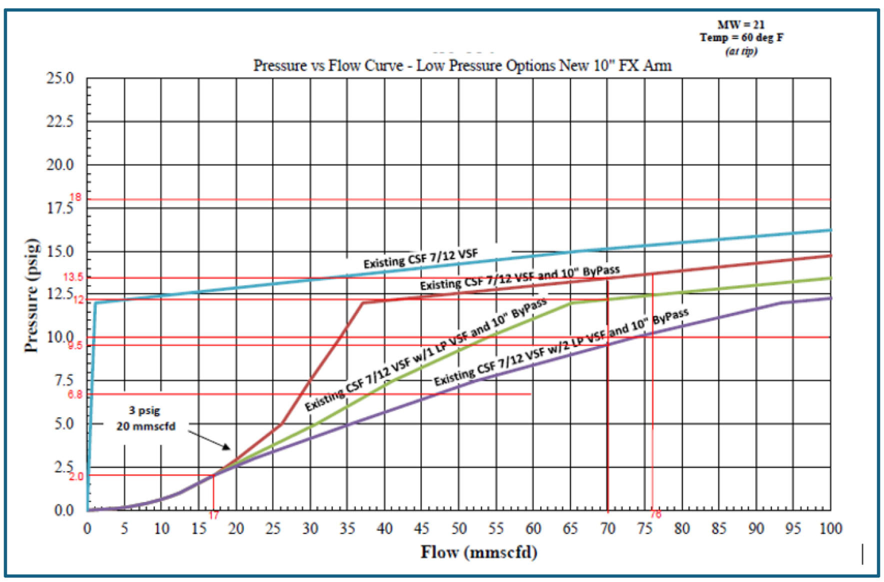


Figure 8—Pressure Vs Flow Curve – with additional low-pressure arm and modification to existing arms[2]

## Results & Discussion

Following operational scenarios, flowrate and corresponding backpressure is studied for each source for each flare header with modified tip design.

**Table 1—Review of backpressure for various operating scenarios**

| Sl. No. | Scenario Description | Flowrate (MMSCFD) | Backpressure (psig) |
|---|---|---|---|
| 1. | During Normal Operation GSU, GDU, VRU, GIC Running | 0.05 | 0.0 |
| 2. | VRU Tripped, all sources routed to Flare and GSU, GDU, GIC Running | 1.8 (Acid Gas) + 15 (Dilution Gas) = 16.8 | 2.0 |
| 3. | VRU, GSU, GDU running, One GIC Train Trip (effect in first 10 min of blowdown) | 70 | 9.5 |
| 4. | VRU, GSU, GDU running, One GIC Train START UP | 5 (FUEL Gas) + 1 (seal gas) = 6 | 0.0 |
| 5. | VRU, GSU, GDU running, One GIC Train START UP and other GIC Train Trips | 6 (from starting train) + 70 (from tripping train) = 76 | 10.0 |
| 6. | VRU, GSU, GIC, GDU running, Depressurization of Associated gas BDV to flare etc. | Approx.1.5 (Controlled riser depressurization) | 0.0 |

The current variable area sonic flare tips in operation are experiencing significant challenges due to high back pressure. This condition poses a risk of facility trips, particularly within low-pressure systems, and necessitates increased purging to maintain operational stability. The proposed modification involves the introduction of low-pressure arms to the flare tip assembly, which is expected to resolve these operational issues effectively. With the modified flare tip, all the current operational issues with low-pressure systems are resolved.

The upgraded flare tip design[1] features a multi-arm configuration equipped with springs to achieve sonic velocity. A variable Mach sonic flare is used for smokeless flaring requirements. The VariMach sonic flare is designed to minimize radiation, utilizes high pressure flare gas to establish a sonic velocity to combust waste gas Smokeless. The spring-actuated variable sonic nozzle allows the exit area to vary with pressure, ensuring constant sonic velocity of waste gas in a wide range of flow with beneficial effects on the smokeless combustion and the lowest purge rate.

This design enhancement ensures that the flare tip can handle varying flow rate and pressures more efficiently, thereby improving overall system performance. Additionally, a retainer cone[1] has been integrated into the center arm of the flare tip. This component plays a critical role in stabilizing the flow and maintaining lower back pressure during low flow / normal operating conditions. This design will ensure that the flare system operates safely and effectively at a turndown flow rate of 20 mmscfd at 1.2 psig, while maintaining the ability to handle higher flow rates i.e. more than 54 mmscfd. The change improves the range of flaring for this Flare tip, making it a feasible solution to meet the desired operation without compromising on quality and safety.

In terms of execution, the option of replacing the entire flare tip with a new design incorporating all the necessary features was thoroughly explored. This approach offers several advantages. Firstly, it significantly minimizes the duration of plant shutdowns, which is crucial for maintaining production continuity and reducing downtime costs. Secondly, it substantially lowers the barge engagement costs associated with offshore works, as the reduced time required for the replacement translates to lower operational expenses.

Overall, the proposed modifications to the flare tip assembly are expected to address the current operational challenges effectively. By implementing these changes, the facility can achieve improved safety, reliability, and cost-efficiency in its flare system operations. designed to minimize radiation, suitable for both offshore and onshore installations, utilizes high

## Conclusion

The upgraded flare tip design[1] with a multi-arm tip and springs to achieve sonic velocity resulted in lower back pressure at normal operating conditions contributing to uninterrupted operation during process upsets avoiding possibility of unit trip. It also allowed retaining sonic capability at a wide range of relief rates.

This new design allowed:

- Low back pressure at normal operating flow reduces the possibility of unit trips.
- Improved back pressure management, offering low back pressure of 2 – 3 psig at lower flow rates (up to 20 MMSCFD, dilution gas to flare), in contrast to the old tip which provides more than 12 psig.
- Ensure that the glycol/amine regeneration system and vapor recovery system operate without disturbances.
- Smooth change-over of the VRU without bypassing safety functions, such as PAHH.
- Retained Sonic capability at higher flow. Also, it lowered the purge requirement for flare based on new design of flare tip.
- **Value Gain**: New flare tip will help to reduce the GHG emission (saving **10 MMSCF of fuel gas** as purge gas thereby reduction of **727 tons of CO2 emission annually**) due to lower purge rates compared to old flare tip.

## References

1. Aereon / Flare Industries: Aereon | Environmental Solutions Company | Oil & Gas Solution Provider
2. GBA-Corona, Inc.: GBA-Corona, Inc. | Flares for the Future